

More matter, with less art.

Hamlet II ii

The quantum mechanics of a single electron in a central potential leads to classification of the one-electron states in terms of four quantum numbers. The individual electrons' spin and orbital angular momenta are coupled in an isolated many-electron ion to give total spin and orbital quantum numbers S and L . Spin-orbit coupling then operates to split the energy levels into a series of J -multiplets, the lowest of which is specified by Hund's rules. Curie-law susceptibility, $\chi = C/T$, is calculated for a general value of J . When placed in a solid, the ion experiences a crystal field due to the charge environment which disturbs the spin-orbit coupling, leaving either S or J as the appropriate quantum number. The crystal field modifies the structure of the lowest M_S or M_J magnetic sublevels which are split by the Zeeman interaction and it introduces single-ion anisotropy.

Atomic physics is concerned with the energy levels of an atom or ion and the possible transitions between them, which are usually in the optical or ultraviolet energy range (1–10 eV). Magnetism is concerned with the energy levels that are occupied at ambient temperature, which usually means only the *ground state*, and its sublevels resulting from interactions with magnetic or electric fields, which produce a splitting of less than 0.1 eV. At ambient temperature $k_B T$ is about 25 meV.

4.1 The hydrogenic atom and angular momentum

The problem of a single electron in a central potential is treated in numerous texts on quantum mechanics and atomic physics. We summarize the results here, because we need to understand the symmetry of the one-electron states – mainly d -states – which are important in magnetism. A hydrogenic atom is composed of a nucleus of charge Ze at the origin and an electron with position labelled by r, θ, ϕ . First, consider a single electron in a central potential $\varphi_e = Ze/4\pi\epsilon_0 r$. The Hamiltonian is

$$\mathcal{H} = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r}. \quad (4.1)$$

Spherical polar coordinates are appropriate for a problem with this symmetry. The operator ∇^2 is then (Appendix C)

$$\nabla^2 = \frac{1}{r^2 \sin \theta} \left[\sin \theta \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin \theta} \frac{\partial^2}{\partial \phi^2} \right]. \quad (4.2)$$

This is rearranged as

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right). \quad (4.3)$$

The term in parentheses which takes account of all the angular variation may be recognized as $-\hat{I}^2/\hbar^2$ (3.15), where $\hat{I}$ is the orbital angular momentum operator.

Schrödinger's equation for the energy levels of the atom is

$$\mathcal{H}\psi_i = \varepsilon_i \psi_i,$$

where ε_i are the energy eigenvalues and ψ_i are the corresponding wave functions. The wave function ψ determines the probability of finding the electron in a small volume dV at $\mathbf{r}$ as $\psi^*(\mathbf{r})\psi(\mathbf{r})dV$, where ψ^* is the complex conjugate of ψ ¹. The Schrödinger equation is therefore:

$$\left[-\frac{\hbar^2}{2m_e} \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{1}{\hbar^2 r^2} \hat{I}^2 \right) - \frac{Ze^2}{4\pi\epsilon_0 r} \right] \psi_i = \varepsilon_i \psi_i. \quad (4.4)$$

Solutions of partial differential equations like (4.4) are obtained by separation of variables; they may be written in the form

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi).$$

Each factor depends on only one of the variables.

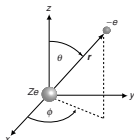
The azimuthal (ϕ -dependent) part of the solution is an eigenfunction of $\hat{I}_z = -i\hbar \partial/\partial \phi$ (3.15). The eigenvalues are $m_\ell \hbar$, with $m_\ell = 0, \pm 1, \pm 2, \dots$ and the corresponding eigenfunctions are $\Phi(\phi) = \exp(im_\ell \phi)$. The polar (θ -dependent) part of the solution is an associated Legendre polynomial $\Theta(\theta) = P_\ell^{m_\ell}(\theta)$, which depends on the angular momentum quantum number ℓ and also on m_ℓ . The quantum number $\ell \geq |m_\ell|$, and $\ell = 0, 1, 2, 3, \dots$ while $m_\ell = 0, \pm 1, \pm 2, \dots, \pm \ell$. There are $(2\ell + 1)$ different values of m_ℓ for a given ℓ .

The product of the azimuthal and polar parts is a spherical harmonic, which depends on two integers ℓ and m_ℓ , such that $\ell \geq 0$ and $|m_\ell| \leq \ell$:

$$Y_\ell^{m_\ell}(\theta, \phi) \propto P_\ell^{m_\ell}(\theta) e^{im_\ell \phi}.$$

The normalized spherical harmonics are listed in Table 4.1. The square of the orbital angular momentum $\hat{I}^2$ has eigenvalues $\ell(\ell + 1)\hbar^2$. Thus the orbital angular momentum has magnitude $\sqrt{\ell(\ell + 1)}\hbar$, and its projection along Oz

¹ The complex conjugate of a complex number is obtained by replacing i by $-i$, where $i = \sqrt{-1}$.



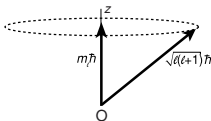
Spherical polar coordinates.

Table 4.1. The normalized spherical harmonics

<i>s</i>	$Y_0^0 = \sqrt{1/4\pi}$	
<i>p</i>	$Y_1^0 = \sqrt{3/4\pi} \cos \theta$	$Y_1^{\pm 1} = \pm \sqrt{3/8\pi} \sin \theta \exp(\pm i\phi)$
<i>d</i>	$Y_2^0 = \sqrt{5/16\pi} (3 \cos^2 \theta - 1)$	$Y_2^{\pm 1} = \pm \sqrt{15/8\pi} \sin \theta \cos \theta \exp(\pm i\phi)$
<i>f</i>	$Y_3^0 = \sqrt{7/16\pi} (5 \cos^3 \theta - 3 \cos \theta)$	$Y_3^{\pm 1} = \pm \sqrt{21/64\pi} (5 \cos^2 \theta - 1) \sin \theta \exp(\pm i\phi)$
<i>s</i>		
<i>p</i>		
<i>d</i>	$Y_2^{\pm 2} = \sqrt{15/32\pi} \sin^2 \theta \exp(\pm 2i\phi)$	
<i>f</i>	$Y_3^{\pm 2} = \sqrt{105/32\pi} \sin^2 \theta \cos \theta \exp(\pm 2i\phi)$	$Y_3^{\pm 3} = \pm \sqrt{35/64\pi} \sin^3 \theta \exp(\pm 3i\phi)$

Figure 4.1

The total angular momentum and its *z* component. The former precesses around *Oz*.



can take any of $(2\ell + 1)$ values from $-\ell\hbar$ to $+\ell\hbar$ (Fig. 4.1). As explained in §3.1.4, the quantities ℓ_z and ℓ^2 can be measured simultaneously because their operators are diagonal matrices which commute.

The *radial part* of the wavefunction $R(r)$ depends on ℓ and n . The latter also takes only integer values, and is known as the principal quantum number; $n > \ell$; hence $\ell = 0, 1, \dots, (n - 1)$. Thus there are n different ℓ values for a given n .

$$R(r) = V_n^\ell(Zr/na_0) \exp[-(Zr/na_0)]$$

where V_n^ℓ are related to the associated Laguerre polynomials; the first one, V_1^0 , equals 1. Here $a_0 = 4\pi\epsilon_0\hbar^2/m_e e^2$ is the Bohr radius (3.8), the basic length scale in atomic physics; $a_0 = 52.92$ pm.

The energy levels of the one-electron atom with a central Coulomb potential $V(r)$ depend on n , but *not* on ℓ or m_ℓ :

$$\varepsilon_n = \frac{-Z^2 m_e e^4}{8\epsilon_0^2 \hbar^2 n^2} = \frac{-Z^2 R_0}{n^2}. \quad (4.5)$$

Here $R_0 = m_e e^4 / 8\epsilon_0^2 \hbar^2$ is the Rydberg, the basic energy scale in atomic physics; $R_0 = 13.61$ eV.

The three quantum numbers n , ℓ , m_ℓ specify a wave function $\psi_i(r, \theta, \phi)$ with a characteristic spatial distribution of electronic charge known as an *orbital*. For historical reasons related to the appearance of lines in atomic spectra, the notation for orbitals is

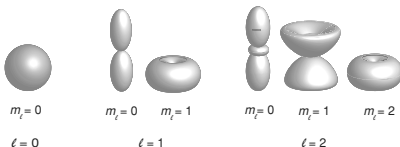
$$nX_{m_\ell}, \quad (4.6)$$

Table 4.2. The hydrogenic orbitals. The number of states per orbital is $2(2\ell + 1)$

	n	ℓ	m_ℓ	m_s	States
1s	1	0	0	$\pm \frac{1}{2}$	2
2s	2	0	0	$\pm \frac{1}{2}$	2
2p	2	1	0, ± 1	$\pm \frac{1}{2}$	6
3s	3	0	0	$\pm \frac{1}{2}$	2
3p	3	1	0, ± 1	$\pm \frac{1}{2}$	6
3d	3	2	0, ± 1 , ± 2	$\pm \frac{1}{2}$	10
4s	4	0	0	$\pm \frac{1}{2}$	2
4p	4	1	0, ± 1	$\pm \frac{1}{2}$	6
4d	4	2	0, ± 1 , ± 2	$\pm \frac{1}{2}$	10
4f	4	3	0, ± 1 , ± 2 , ± 3	$\pm \frac{1}{2}$	14

Figure 4.2

Some hydrogenic orbitals specified by their ℓ and m_ℓ values.



where $x = s, p, d, f$ stands for $\ell = 0, 1, 2, 3$. Each orbital can accommodate up to two electrons with spin $m_s = \pm \frac{1}{2}$. No two electrons can be in a state with the same four quantum numbers, which is a way of stating the Pauli exclusion principle. It is forbidden for two electrons to occupy the same quantum state because they are fermions.

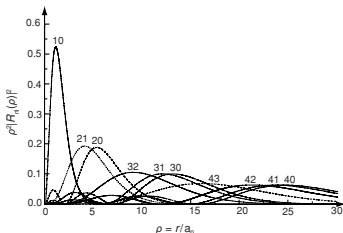
The possible hydrogenic orbitals are enumerated in Table 4.2. The angular parts of the wave functions are the spherical harmonics illustrated in Fig. 4.2, and the radial functions are plotted in Fig. 4.3. Note that the radial functions have between 0 and $n - 1$ nodes, depending on the value of ℓ . The number of different orbitals for an ℓ -shell is $2\ell + 1$; the s, p, d, f shells having $\ell = 0, 1, 2, 3$ contain 1, 3, 5, 7 orbitals respectively.

4.2 The many-electron atom

When more than one electron is present, mutual Coulomb repulsion terms between the electrons $e^2/4\pi\epsilon_0 r_{ij}$ must be added into (4.1). The Hamiltonian

Figure 4.3

Radial probability distributions deduced from the radial parts of the wave function $R(r)$. The curves are labelled with the values of n and ℓ .



becomes

$$\mathcal{H}_0 = \sum_i [-(\hbar^2/2m_e)\nabla^2 - Ze^2/4\pi\epsilon_0 r_i] + \sum_{i<j} e^2/4\pi\epsilon_0 r_{ij}. \quad (4.7)$$

The sum over many pairs of interacting particles makes this an intractable analytical problem.² An appropriate way of dealing with the extra Coulomb interactions is to suppose that each electron experiences the central potential of some different, spherically symmetric charge distribution. The potential with many electrons is no longer a simple Coulomb well and the degeneracy of the energy of electrons with different ℓ -values but the same principal quantum number n is lifted. The $4s$ shell, for example, then turns out to be lower in energy than the $3d$ shell, the energy change depends on filling. The sense in which the shells are filled defines the shape of the periodic table. The potentials can be determined self-consistently. This is known as the Hartree-Fock approximation.

When several electrons are present on the same atom, at most two of them, with opposite spin, can occupy the same orbital. The ions of interest in magnetism generally follow the L-S coupling scheme, where individual spin and orbital angular momenta add³ to give resultant quantum numbers (here S and $L \geq 0$):

$$S = \sum s_i, \quad M_S = \sum m_{si}, \quad L = \sum \ell_i, \quad M_L = \sum m_{\ell i}.$$

n	1	2	3	4	5	6
1	1s	2s	3s	4s	5s	6s
2		2p	3p	4p	5p	6p
3			3d	4d	5d	6d
4				4f	5f	6f
5					5g	6g

Sequence of shell filling for a many electron atom.

² There is a very complicated analytical solution to the three-body problem, but there are usually many more than three particles involved in atoms.

³ When spin-orbit coupling is very strong, as it is in the actinides, it is appropriate to first couple ℓ_i and s_i for each electron, to form j_i , and then to couple these total angular momenta. This is the $j-j$ coupling scheme.

Table 4.3. Example of the six-electron carbon atom; $1s^2 2s^2 2p^2$

1s	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓
2s	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓
2p ₋₁	↑↓	↑	↑	↓	↓	↑	↑			↓	↓				
2p ₀		↑	↓	↑	↓			↑↓			↓	↑	↓	↑	
2p ₁						↑	↓		↑	↓			↑	↓	↑↓
M _L	2	1	1	1	1	0	0	0	0	0	-1	-1	-1	-1	-2
M _S	0	-1	0	0	+1	-1	0	0	0	+1	+1	0	0	-1	0

Table 4.4. Terms for the carbon atom.

Term	L	S	(M _L , M _S)
¹ S	0	0	(0,0)
³ P	1	1	(1, 1)(1, 0)(1, -1)(0, 1)(0, 0)(0, -1)(-1, 1)(-1, 0)(-1, -1)
¹ D	2	0	(2, 0)(1, 0)(0, 0)(-1, 0)(-2, 0)

Consider, for example, the carbon atom with its six electrons. The electronic configuration is $1s^2 2s^2 2p^2$. Each s -shell has no alternative but to accommodate the pair of electrons with opposite spin. However, there are 15 ways of accommodating two electrons in the $2p$ -orbitals. The possibilities are listed in Table 4.3, a subscript denotes the m_l value.

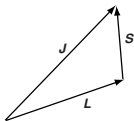
The 15 states are grouped into three terms. This is done by considering the number of states with a particular combination of M_L and M_S , and decomposing them into blocks, as follows:

$M_S \backslash M_L$	-2	-1	0	1	2	$L = 1, S = 1$	$L = 2, S = 0$	$L = S = 0$
-1	—	1	1	1	—	1	1	1
0	1	2	3	2	1	1	1	1
1	—	1	1	1	—	1	1	1

and identifying each block as a term. Terms for carbon are listed in Table 4.4. The convention is to denote $L = 0, 1, 2, 3, 4, 5$ by S, P, D, F, G, H , and then indicate the spin multiplicity $2S + 1$ by a left superscript ^{2S+1}X . The energy splitting of the terms is of order 1 eV, so the allowed $\Delta L = \pm 1$ transitions to excited states require optical excitation. The final step is to introduce spin-orbit interaction which couples L and S together to form J . Each term gives rise to several J -states, with $|L - S| \leq J \leq L + S$. There are either $(2S + 1)$ or $(2L + 1)$ of them, depending on whether L or S is smaller. The different J -states of a term are known as multiplets. The general notation for multiplets shows the J -value as a subscript on the term, thus

$$^{2S+1}X_J.$$

The ground state is normally the only one of the $(2J + 1)$ M_J -states occupied at room temperature; it determines the magnetic properties of the atom.



Addition of spin and orbital angular momentum in the vector model: $\mathbf{J} = \mathbf{L} + \mathbf{S}$.

Hund provided an *empirical* prescription for determining the lowest-energy state of a multielectron atom or ion; Hund's rules are:

- (1) First maximize S for the configuration.
- (2) Then maximize L consistent with S .
- (3) Finally couple L and S to form J : $J = L - S$ if the shell is less than half full, and $J = L + S$ if the shell is more than half full. When the shell is exactly half full, $L = 0$ and $J = S$.

The justification for the first rule is that electrons minimize their Coulomb interaction by avoiding each other, which is best achieved if they can occupy different orbitals. To this end, intra-atomic exchange tends to align the spins of different orbitals parallel. The second rule means that the electrons orbit in the same sense whenever possible. The third and weakest rule is a consequence of the sign of the spin-orbit coupling. Hund's rules only predict the *ground state*; they tell us nothing about the position and order of the excited states.

In our example of carbon, Hund's rules give $S = 1$, $L = 1$ and $J = 0$, so $M_J = 0$, and the ground state of the carbon atom is actually *nonmagnetic*, thanks to spin-orbit coupling. Free atoms of other elements in the same column of the magnetic periodic table are likewise nonmagnetic, as well as s^2 atoms, p^6 atoms and a few others (Table A).

More examples of the use of Hund's rules to determine the ground-state multiplet of some common magnetic ions are:

Fe^{3+}	$3d^5$	$\uparrow\uparrow\uparrow\uparrow\uparrow$	ooooo	
$S = 5/2$		$L = 0$	$J = 5/2$	${}^6S_{5/2}$
Ni^{2+}	$3d^8$	$\uparrow\uparrow\uparrow\uparrow\uparrow$	$\downarrow\downarrow\downarrow$ oo	
$S = 1$		$L = 3$	$J = 4$	3F_4
Nd^{3+}	$4f^3$	$\uparrow\uparrow\uparrow$	ooooooo	
$S = 3/2$		$L = 6$	$J = 9/2$	${}^4I_{9/2}$
Dy^{3+}	$4f^9$	$\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow$	$\downarrow\downarrow$ ooooo	
$S = 5/2$		$L = 5$	$J = 15/2$	${}^6H_{15/2}$

The symbols $\uparrow$, $\downarrow$ and o show whether a $3d$ - or $4f$ -orbital is occupied by a $\uparrow$ or a $\downarrow$ electron, or is unoccupied. Values of L , S and J for the whole series of $3d^n$ and $4f^n$ ions are plotted in Fig. 4.4. We saw already in Table A that most free atoms possess a magnetic moment.

4.2.1 Spin-orbit coupling

We return to the relatively weak relativistic interaction responsible for Hund's third rule. The spin-orbit interaction is the origin of many of the most

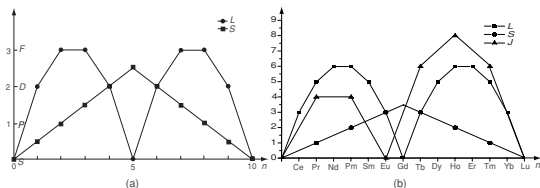


Figure 4.4

(a) L and S for the series of $3d^n$ ions; (b) L , S and J for the series of trivalent $4f$ ions.

interesting phenomena in magnetism, including magnetocrystalline anisotropy, magnetostriction, anisotropic magnetoresistance and the anomalous planar and spin Hall effect.

In the multielectron atom, the single-electron spin-orbit coupling we encountered in §3.1.3 becomes

$$\mathcal{H}_{so} = (\Lambda/\hbar^2) \hat{\mathbf{L}} \cdot \hat{\mathbf{S}}.$$

As Hund's third rule testifies, Λ is positive for the first half of the $3d$ or $4f$ series and negative for the second half. The coupling becomes large in heavy elements where the energy of the orbiting electrons is a significant fraction of their rest-mass energy $m_e c^2$. The constant Λ is related to the one-electron coupling constant λ of (3.13) by $\Lambda = \pm \lambda/2S$ for the first and second halves of the series (Table 4.5). Since $\mathbf{J} = \mathbf{L} + \mathbf{S}$, the identity $J^2 = L^2 + S^2 + 2\mathbf{L} \cdot \mathbf{S}$ can be used to evaluate $\mathcal{H}_{so}$, as the eigenvalues of J^2 , L^2 and S^2 are all known. Turning on the spin-orbit coupling, the atomic states in the L – S coupling scheme are labelled by (L, S, J, M_J) , where M_J is the total magnetic quantum number.

4.2.2 The Zeeman interaction

By analogy with (3.20), the magnetic moment of an atom is represented by the operator

$$\hat{\mathbf{m}} = -(\mu_B/\hbar)(\hat{\mathbf{L}} + 2\hat{\mathbf{S}}). \quad (4.8)$$

The Zeeman Hamiltonian for the magnetic moment in a field $\mathbf{B}$ is

$$\mathcal{H}_Z = (\mu_B/\hbar)(\hat{\mathbf{L}} + 2\hat{\mathbf{S}}) \cdot \mathbf{B}. \quad (4.9)$$

When $\mathbf{B}$ is applied along Oz , the Hamiltonian becomes $(\mu_B/\hbar)(\hat{L}_z + 2\hat{S}_z)B$.

Next we define the Landé g -factor for the multielectron atom or ion as the ratio of the component of magnetic moment along $\mathbf{J}$ in units of μ_B to the magnitude

Table 4.5. Spin-orbit coupling constants for ions in the $3d$ and $4f$ series, in kelvin. $\Delta\epsilon$ is the energy of the first excited multiplet

	Ion	λ	Λ	$\Delta\epsilon$
$3d^1$	Sc ²⁺	124	124	310
$3d^2$	Ti ²⁺	176	88	264
$3d^3$	V ²⁺	246	82	205
$3d^4$	Cr ²⁺	340	85	85
$3d^6$	Fe ²⁺	656	-164	656
$3d^7$	Co ²⁺	818	-272	1224
$3d^8$	Ni ²⁺	987	-494	3948
$4f^1$	Ce ³⁺	920	920	3220
$4f^2$	Pr ³⁺	1080	540	2700
$4f^3$	Nd ³⁺	1290	430	2365
$4f^4$	Pm ³⁺	1540	380	1900
$4f^5$	Sm ³⁺	1730	350	1225
$4f^6$	Eu ³⁺	1950	330	330
$4f^8$	Tb ³⁺	2450	-410	2460
$4f^9$	Dy ³⁺	2730	-550	4125
$4f^{10}$	Ho ³⁺	3110	-780	6240
$4f^{11}$	Er ³⁺	3510	-1170	8775
$4f^{12}$	Tm ³⁺	3800	-1900	11400
$4f^{13}$	Yb ³⁺	4140	-4140	14490

of the angular momentum, in units of $\hbar$. The g -factor is a dimensionless version of the gyromagnetic ratio γ (3.5).

Thus, in vector notation

$$g = -(\mathbf{m} \cdot \mathbf{J} / \mu_B) / (|\mathbf{J}|^2 / \hbar) = -\mathbf{m} \cdot \mathbf{J} / J(J+1)\mu_B\hbar,$$

but

$$\begin{aligned} \mathbf{m} \cdot \mathbf{J} &= -(\mu_B/\hbar)[(\mathbf{L} + 2\mathbf{S}) \cdot (\mathbf{L} + \mathbf{S})] \\ &= -(\mu_B/\hbar)[(\mathbf{L}^2 + 3\mathbf{L} \cdot \mathbf{S} + 2\mathbf{S}^2)] \text{ since } \mathbf{L} \text{ and } \mathbf{S} \text{ commute} \\ &= -(\mu_B/\hbar)[\mathbf{L}^2 + 2\mathbf{S}^2 + (3/2)(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2)] \\ &= -(\mu_B\hbar)[(3/2)J(J+1) - (1/2)L(L+1) + (1/2)S(S+1)]. \end{aligned}$$

The expression for the g -factor is therefore

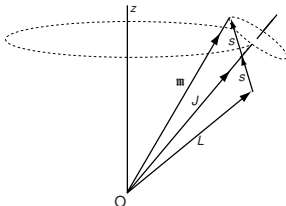
$$g = \frac{3}{2} + \{S(S+1) - L(L+1)\}/2J(J+1). \quad (4.10)$$

The Landé g -factor is also the ratio of the z components of magnetic moment (in units of μ_B) and angular momentum (in units of $\hbar$). From the vector model, Fig. 4.5, where the magnetic moment precesses rapidly about $\mathbf{J}$, the ratio

Figure 4.5

The vector model of the atom, showing both the magnetic moment and the angular momentum.

$\mathbf{m}$ precesses rapidly around $\mathbf{J}$, and the time-averaged moment then precesses around Oz .



$m_z/J_z = \mathbf{m} \cdot \mathbf{J} / J^2 = -g\mu_B/\hbar$, so we can write the projected/average $\mathbf{m}$ in terms of the g -factor as

$$\mathbf{m} = -(g\mu_B/\hbar)\mathbf{J}.$$

The level with $M_J = -J$ is lowest.

The Zeeman energy in a magnetic field $\mathbf{B}$ applied along Oz is $\varepsilon_Z = -\mathbf{m}_z \cdot \mathbf{B}$. This is $-(m_z/J_z)J_z B = (g\mu_B/\hbar)J_z B$. Hence the Zeeman energy is

$$\varepsilon_Z = g\mu_B M_J B.$$

Note the magnitude of the energies involved here. The splitting of two adjacent Zeeman energy levels is $g\mu_B B$, which is of order 1 K for a field of 1 T. Hundreds of tesla would be needed to establish a significant population surplus in the $M_J = -J$ ground state at room temperature. No such steady field has ever been produced in the laboratory. Matters are easier in the liquid helium temperature range, where normal laboratory fields of a few tesla will suffice to establish a preponderant population of the ground state, and saturate the magnetization of the ion. The entropy S of the ion is $k_B \ln \Omega$, where Ω is the number of configurations. When the $2J + 1$ M_J configurations are equally populated, $S = k_B \ln(2J + 1)$, but if only one M_J sublevel is occupied, $S = 0$.

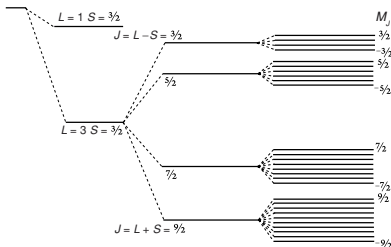
An example to summarize the discussion up to this point is given in Fig. 4.6, where the energy levels of a free ion, $\text{Co}^{2+} 3d^7$, are shown. According to Hund's rules, the $J = \frac{9}{2}$ multiplet is the ground state.

4.3 Paramagnetism

We now examine in more detail the response of a localized magnetic moment $\mathbf{m}$ to an applied magnetic field $\mathbf{H} = \mathbf{B}/\mu_0$ in free space. The ϵ is dropped because

Figure 4.6

The energy levels for a free ion with electronic configuration $3d^7$: Co^{2+}
 $S = \frac{3}{2}$, $L = 3$, $J = \frac{9}{2}$; $g = \frac{6}{5}$.



there is no distinction between applied field and internal field for a single atom or particle. It is instructive to look first at the general quantum case where the moment $\mathbf{m} = -g\mu_B \mathbf{J}/\hbar$ can make any of $2J + 1$ polar angles with the applied field, and then at the theory for classical magnetic moments which can adopt any orientation whatsoever with respect to the applied field. We have already presented the theory for a spin- $\frac{1}{2}$ quantum system in §3.2.4, where the moment $\mathbf{m} = -g\mu_B \mathbf{S}/\hbar$ can have only one of two projections along the applied field. The classical and extreme quantum limits correspond to $J \rightarrow \infty$ and $J = \frac{1}{2}$, respectively.

4.3.1 Brillouin theory

The general expression for the thermodynamic average value of any quantity q is $\langle q \rangle = \text{Tr}[q_i \exp(-\varepsilon_i/k_B T)]/\mathcal{Z}$, where $\mathcal{Z} = \text{Tr} \exp(-\varepsilon_i/k_B T)$ is the partition function. The trace (Tr) is the sum over the i energy states, and q_i is the value of q in the i th one. This amounts to taking an average over the i states, weighted by their Boltzmann populations. $\mathcal{Z}$ is the normalization factor. Evaluating the thermodynamic average $\langle \mathbf{m} \rangle$,

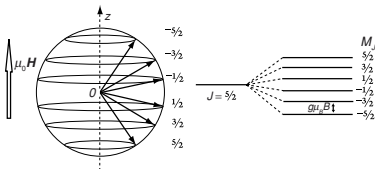
$$\langle \mathbf{m} \rangle = \frac{\sum_i \mathbf{m}_i \exp(-\varepsilon_i/k_B T)}{\sum_i \exp(-\varepsilon_i/k_B T)}. \quad (4.11)$$

The extreme quantum limit is the case $J = 1/2$. Usually this arises when $S = 1/2$, $L = 0$. There are only two energy levels and two orientations of the moment relative to the applied field (which, as usual, is along Oz), corresponding to the $|\uparrow\rangle$ and $|\downarrow\rangle$ states. Then (4.11) for the z component of $\mathbf{m}$ reduces to

$$\langle m_z \rangle = g\mu_B J \tanh x, \quad (4.12)$$

Figure 4.7

A $J = \frac{5}{2}$ quantum moment in an applied field. The Zeeman shift is shown on the right.



where x is the dimensionless ratio of the Zeeman energy to the thermal energy.

$$x = \mu_0 g \mu_B M_J H / k_B T. \quad (4.13)$$

When $J = S = \frac{1}{2}$, $g = 2$, this becomes $\langle m_z \rangle = m_z \tanh x$, with $m_z = \mu_B$ and $x = \mu_0 \mu_B H / k_B T$, as seen in §3.2.4. In small fields, $\tanh x \approx x$, and if there are n atoms per unit volume, the susceptibility $n \langle m_z \rangle / H$ is

$$\chi = \mu_0 n g^2 \mu_B^2 J^2 / k_B T. \quad (4.14)$$

Equation (4.14) reduces to (3.34) for $J = 1/2$, $g = 2$. Many two-level systems in physics are treated using this theory, by assigning them a pseudospin $S = \frac{1}{2}$; the two levels correspond to $M_x = \pm \frac{1}{2}$, and their separation is $2\mu_0 \mu_B H$.

The general quantum case was treated by Léon Brillouin; m is $-g\mu_B J/\hbar$, and x is $\mu_0 g \mu_B M_J H / k_B T$. There are now $2J + 1$ energy levels $\varepsilon_i = +\mu_0 g \mu_B M_J H$, with moments $m_{zi} = -g\mu_B M_J$, where $M_J = J, J - 1, J - 2, \dots, -J$. The sums in (4.11) each have $2J + 1$ terms. The example of $J = \frac{5}{2}$ is illustrated in Fig. 4.7.

To calculate the susceptibility, we take the limit $x \ll 1$. Susceptibility is the initial slope of the magnetization curve. The exponentials in (4.11) are expanded as $\exp(x) \approx 1 + x + \dots$. Hence, keeping only the leading terms,

$$\langle m_z \rangle = \frac{\sum_{-J}^J -g\mu_B M_J (1 - \mu_0 g \mu_B M_J H / k_B T)}{\sum_{-J}^J (1 - \mu_0 g \mu_B M_J H / k_B T)}.$$

Using the identities

$$\sum_{-J}^J 1 = 2J + 1, \quad \sum_{-J}^J M_J = 0, \quad \sum_{-J}^J M_J^2 = J(J + 1)(2J + 1)/3,$$

we find $\langle m_z \rangle = \mu_0 g^2 \mu_B^2 J(J + 1)H / 3k_B T$. The susceptibility is $n \langle m \rangle / H$, thus the general form of Curie's law is $\chi = C/T$, where

$$C = \frac{\mu_0 n g^2 \mu_B^2 J(J + 1)}{3k_B}. \quad (4.15)$$

To make a connection with (3.34), the susceptibility is written in terms of an effective moment $m_{\text{eff}} = g\mu_B\sqrt{J(J+1)}$ (or an effective Bohr magneton number $p_{\text{eff}} = g\sqrt{J(J+1)}$) as

$$\chi = \frac{\mu_0 n m_{\text{eff}}^2}{3k_B T}.$$

The susceptibility depends on the square of the magnitude or 'length' of m , and not on its projection m_z . The Curie constant is

$$C = \mu_0 n m_{\text{eff}}^2 / 3k_B. \quad (4.16)$$

A typical value of C obtained for $n = 6 \times 10^{28} \text{ m}^{-3}$, $g = 2$ and $J = 1$ is 1.3 K. The molar Curie constant C_{mol} , obtained by setting $n = N_A$, is a useful quantity. Its numerical value is

$$C_{\text{mol}} = 1.571 \times 10^{-6} p_{\text{eff}}^2.$$

In order to calculate the complete magnetization curve, we set $y = \mu_0 g \mu_B H / k_B T$ and then use (4.11) and the result $d(\ln z)/dy = (1/z)dz/dy$ to write the thermodynamic average $\langle m_z \rangle = \langle -g \mu_B M_J \rangle$ as

$$\langle m_z \rangle = g \mu_B \frac{\partial}{\partial y} \left[\ln \sum_{-J}^J \exp(-M_J y) \right].$$

The sum over the energy levels must be evaluated; it can be written as $\exp(Jy)[1 + r + r^2 + \dots + r^{2J}]$, where $r = \exp(-y)$. The sum of the geometric progression is $(r^{2J+1} - 1)/(r - 1)$. Therefore, multiplying top and bottom by $\exp(y/2)$,

$$\begin{aligned} \sum_{-J}^J \exp(M_J y) &= \{\exp[-(2J+1)y] - 1\} \exp(Jy) / [\exp(-y) - 1] \\ &= \sinh[(2J+1)y/2] / \sinh(y/2). \end{aligned}$$

Hence

$$\begin{aligned} \langle m_z \rangle &= g \mu_B \frac{\partial}{\partial y} \ln[\sinh((2J+1)y/2)] / [\sinh(y/2)] \\ &= (g \mu_B / 2) [(2J+1) \coth((2J+1)y/2) - \coth(y/2)]. \end{aligned}$$

Setting $x = Jy$ we obtain finally

$$\langle m_z \rangle = m_0 \left\{ \frac{2J+1}{2J} \coth \frac{2J+1}{2J} x - \frac{1}{2J} \coth \frac{x}{2J} \right\}, \quad (4.17)$$

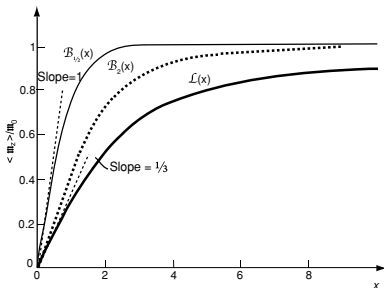
where $m_0 = g \mu_B J$ is the maximum magnitude of the moment and the quantity in braces is the Brillouin function $B_J(x)$:

$$\langle m_z \rangle = m_0 B_J(x). \quad (4.18)$$

It reduces to the Langevin function in the limit $J \rightarrow \infty$ and to $\tanh x$ when $J = 1/2$ and $g = 2$ (4.12), Fig. 4.8.

Figure 4.8

Comparison of the Brillouin functions for $J = \frac{1}{2}$ and $J = 2$ with the Langevin function, which is the classical limit. The slope at the origin of the Langevin function is $\frac{1}{3}$.

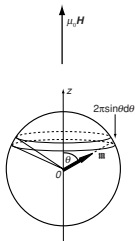


In the small- x limit,

$$B_J(x) \approx \frac{(J+1)}{3J}x - \frac{[(J+1)^2 + J^2](J+1)}{90J^2}x^3 + \dots \quad (4.19)$$

The leading term gives the Curie law susceptibility $\chi = g^2 \mu_0 n J(J+1) \mu_B^2 / 3k_B T = C/T$.

The theory of localized magnetism gives an excellent account of magnetically dilute $3d$ and $4f$ salts, where the magnetic moments do not interact with each other, for example in alums such as $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, where Cr^{3+} ions are well separated by sulphate ions and water of crystallization. Except in large fields or at very low temperatures, the $M(H)$ response is linear. The excellence of the theory is illustrated by the fact that all the data for quite different temperatures superimpose on a single Brillouin curve when plotted as a function of $x \sim H/T$, as shown in Fig. 4.9.



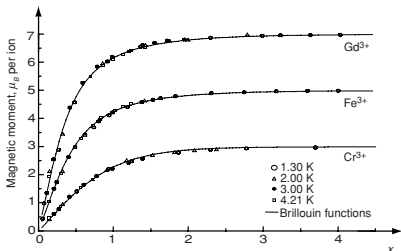
A classical moment in an applied field.

4.3.2 Langevin theory

The classical theory of paramagnetism, which is the limit of quantum theory when $J \rightarrow \infty$, was worked out in 1905 by Paul Langevin. Colloidal ferromagnetic nanoparticles suspended in a liquid or small grains of ferromagnetic minerals, usually magnetite, dispersed in a rock are examples of systems where the classical theory is expected to apply. Each atom or particle has a macroscopic moment m which can take any orientation relative to the field applied in

Figure 4.9

Reduced magnetization curves for three paramagnetic salts, compared with Brillouin theory predictions. (W. E. Henry, *Phys. Rev.* 88, 559 (1952)).



the z -direction $\mathbf{H} = \mathbf{B}/\mu_0$. The Zeeman energy (2.73) $\varepsilon = -\mathbf{m} \cdot \mathbf{B}$ gives

$$\varepsilon(\theta) = -\mu_0 m H \cos \theta.$$

The probability $P(\theta)$ that the moment makes an angle θ with Oz is the product of a Boltzmann factor $\exp[-\varepsilon(\theta)/k_B T]$ and a geometric factor $2\pi \sin \theta$. Hence $P(\theta) = \kappa 2\pi \sin \theta \exp(\mu_0 m H \cos \theta / k_B T)$, where the constant κ is determined by the normalization condition $\int_0^\pi P(\theta) d\theta = N$. Here N is the number of atoms or particles in the system. Then

$$\langle m_z \rangle = \frac{\int_0^\pi m \cos \theta P(\theta) d\theta}{\int_0^\pi P(\theta) d\theta}. \quad (4.20)$$

To evaluate the integrals, let $a = \cos \theta$, $da = -\sin \theta d\theta$ and define the dimensionless ratio of magnetic to thermal energy $x = \mu_0 m H / k_B T$. The result is $\langle m_z \rangle = m \mathcal{L}(x)$, where

$$\mathcal{L}(x) = (\coth x - 1/x). \quad (4.21)$$

The quantity $\mathcal{L}(x)$, known as the Langevin function, is plotted in Fig. 4.8. The leading terms in the expansion when x is small are $\mathcal{L}(x) = x/3 - x^3/45 + 2x^5/945 - \dots$. At low fields or high temperatures, $\mathcal{L}(x) \approx x/3$, which is just $\frac{1}{3}$ of the value in extreme quantum case $J = \frac{1}{2}$. The susceptibility of an ensemble of n moments per cubic metre is $\chi = n \langle m_z \rangle / H$, hence

$$\chi = \mu_0 n m^2 / 3 k_B T. \quad (4.22)$$

This is the classical form of the famous Curie law $\chi = C/T$, where $C = \mu_0 n m^2 / 3 k_B$ is the Curie constant.

At high fields, $x \gg 1$, the moments are aligned and the magnetization saturates. The high-field limit of the Langevin function is $\mathcal{L}(x) \approx 1 - 1/x$.

4.3.3 Van Vleck susceptibility

When an ion has low-lying excited states, they can contribute to the susceptibility in two ways. They may be Zeeman split and thermally populated as usual. Otherwise, the applied field may mix some of the excited-state character into the ground state. This produces a term known as van Vleck paramagnetism or temperature-independent paramagnetism (TIP). It is especially important for Eu^{3+} , which has a nonmagnetic $J = 0$ ground state, but a low-lying $J = 1$ excited state at 330 K (Table 4.5) which contributes a paramagnetic susceptibility. The other rare-earth ion for which it may be necessary to consider the excited J -multiplets is Sm^{3+} , which has a $J = \frac{5}{2}$ ground state and a low-lying $J = \frac{7}{2}$ excited state at 1225 K. Among the actinides, Bk^{5+} and Cm^{4+} have a localized $5f^6$ configuration, with a $J = 0$ ground state.

According to perturbation theory, the correction to the energy of the ion is given by the Hamiltonian (4.9)

$$\varepsilon_1 = \left[\frac{\mu_B}{\hbar} \right]^2 \frac{|\langle g | (\hat{\mathbf{L}} + 2\hat{\mathbf{S}}) \cdot \mathbf{B} | e \rangle|^2}{\Delta\varepsilon}, \quad (4.23)$$

where g and e represent the ground and excited states, respectively, and $\Delta\varepsilon$ is their energy separation. From the expression for the Gibbs free energy of a paramagnet in an applied field, $-\frac{1}{2}BM$ with $M = \chi B/\mu_0$, it follows that $\chi = 2n\mu_0\varepsilon_1/B^2$, where n is the number of ions per unit volume. Hence the susceptibility depends on the off-diagonal matrix elements of the magnetic moment:

$$\chi = 2n\mu_0 \left[\frac{\mu_B}{\hbar} \right]^2 \frac{|\langle g | \hat{\mathbf{L}}_z + 2\hat{\mathbf{S}}_z | e \rangle|^2}{\Delta\varepsilon}. \quad (4.24)$$

4.3.4 Adiabatic demagnetization

A remarkable application of the Curie-law paramagnetism of dilute salts is the achievement of temperatures in the millikelvin range by adiabatic demagnetization.

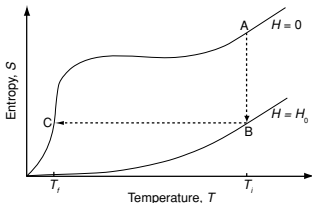
In zero applied field, the energy levels of each ion are $(2J + 1)$ -fold degenerate. The number of possible configurations for N ions in the sample is $\Omega = (2J + 1)^N$. The corresponding entropy is $S = k_B \ln \Omega$ or

$$S = R \ln(2J + 1), \quad (4.25)$$

per mole, where $R = N_A k_B$ is the gas constant, 8.315 J mol^{-1} . The entropy is reduced by applying a magnetic field, which leads to preferential occupation of the energy levels with more negative M_J values, according to the Boltzmann probability $\exp(-g\mu_B M_J \mu_0 H / k_B T)$. At zero temperature, all the ions are in the $M_J = -J$ state. There is only one configuration for each one, so $\Omega = 1^N$ and $S = 0$. Increasing temperature leads to absorption of heat by the

Figure 4.10

Cooling by adiabatic demagnetization. The two curves are $S(T)$ for zero field, $H = 0$, and an applied field, $H = H_0$. The material is first magnetized isothermally at T_i by the applied field H_0 (AB), and then cooled adiabatically to T_f when the field H is reduced almost to zero (BC).



system and population of the higher Zeeman levels, which increases the entropy. Increasing field favours preferential population of the lower Zeeman levels, which decreases the entropy. Since the population of the energy levels depends solely on the Boltzmann factor, entropy is a monotonically decreasing function of H/T . The entropy can be calculated from the Helmholtz free energy $F = -Nk_B T \ln Z$, where Z is the partition function, using the thermodynamic relation $S = -(\partial F / \partial T)_H$.

An adiabatic process is one where there is no exchange of heat between a system and its surroundings. Since $\delta Q = T \delta S$, entropy is conserved in such a process. Adiabatic demagnetization is usually carried out in a cryostat, where, for example, a pill of cerium magnesium nitrate is initially in thermal contact via helium exchange gas with a bath of pumped liquid helium at 1.2 K or liquid ^3He at 0.3 K. The first step (Fig. 4.10) is isothermal magnetization in a field of several teslas, generated by a superconducting magnet. The second step is to cut the thermal contact with the bath by evacuating the exchange gas, and reduce the field to zero. During the adiabatic demagnetization, there is no change of entropy $S(H/T)$ so it follows that

$$\frac{H_i}{T_i} = \frac{H_f}{T_f}, \quad (4.26)$$

where i and f denote the initial and final states. If H_f were really zero, the procedure would attain the absolute zero of temperature! Unfortunately, there is always some residual field due to the dipole-dipole interaction of the ionic moments. These stray fields are of order a millitesla in dilute salts, so the best temperatures that can be reached are in the millikelvin range.

Microkelvin temperatures are attainable by adding a second stage where adiabatic demagnetization is applied to the nuclei. Nuclear magnetic moments are three orders of magnitude smaller than electronic ones, so the stray fields are proportionately less, but the angular momentum $I\hbar$ and the entropy in the disordered state, $R \ln(2I + 1)$ per mole, are similar. Copper, which has two

Table 4.6. The $4f$ ions. The paramagnetic moment m_{eff} and the saturation moment m_0 are in units of μ_B

$4f^n$	S	L	J	g	$m_0 = gJ$	$m_{eff} = g\sqrt{J(J+1)}$	m_{eff}^{exp}
1 Ce^{3+}	$\frac{1}{2}$	3	$\frac{5}{2}$	$\frac{6}{7}$	2.14	2.54	2.5
2 Pr^{3+}	1	5	4	$\frac{4}{5}$	3.20	3.58	3.5
3 Nd^{3+}	$\frac{3}{2}$	6	$\frac{9}{2}$	$\frac{8}{11}$	3.27	3.52	3.4
4 Pm^{3+}	2	6	4	$\frac{5}{3}$	2.40	2.68	
5 Sm^{3+}	$\frac{5}{2}$	5	$\frac{5}{2}$	$\frac{3}{2}$	0.71	0.85	1.7
6 Eu^{3+}	3	3	0	0	0	0	3.4
7 Gd^{3+}	$\frac{7}{2}$	0	$\frac{7}{2}$	2	7.0	7.94	8.9
8 Tb^{3+}	3	3	6	$\frac{3}{2}$	9.0	9.72	9.8
9 Dy^{3+}	$\frac{4}{2}$	5	$\frac{15}{2}$	$\frac{4}{3}$	10.0	10.65	10.6
10 Ho^{3+}	2	6	8	$\frac{15}{4}$	10.0	10.61	10.4
11 Er^{3+}	$\frac{3}{2}$	6	$\frac{15}{2}$	$\frac{12}{5}$	9.0	9.58	9.5
12 Tm^{3+}	1	5	6	$\frac{7}{3}$	7.0	7.56	7.6
13 Yb^{3+}	$\frac{1}{2}$	3	$\frac{7}{2}$	$\frac{8}{7}$	4.0	4.53	4.5

isotopes each with $I = 3/2$, is often used. It is first cooled to the millikelvin range where the nuclei are polarized in an applied field of a few tesla, and then demagnetized adiabatically.

4.4 Ions in solids; crystal-field interactions

In practice, it should be possible to deduce m_0 , the maximum value of m_z , by saturating the magnetization at low temperature, thereby obtaining the value of $gJ\mu_B$. The magnitude of the moment can be deduced from the paramagnetic susceptibility, which yields $m_{eff}^2 = g^2\mu_B^2J(J+1)$. In practice, these measurements are made on magnetically dilute salts, not free ions. Results of these measurements are summarized in the last columns of Tables 4.6 and 4.7, where m_{eff} , listed in units of Bohr magnetons, is just the effective Bohr magneton number p_{eff} .

From data on the $4f$ ions in Table 4.6 it is seen that the experimentally measured m_{eff} corresponds to $g\mu_B\sqrt{J(J+1)}$, except for radioactive Pm, for which it has never been measured, and Sm^{3+} and Eu^{3+} , where there is an additional van Vleck contribution to the susceptibility from excited multiplets. J is the good quantum number for the rare-earth series. However, Table 4.7 tells a different story for the $3d$ ions. There is a large discrepancy between $gJ\mu_B$ and m_0 for all except the $3d^5$ ions, for which Hund's second rule gives $L = 0$ and $J = S$. In fact, $m_0 \approx gS\mu_B$ and $m_{eff} \approx g\mu_B\sqrt{S(S+1)}$. It appears that the magnetism of the $3d$ series is due to the spin moment, with little or no orbital contribution. S is the good quantum number for the $3d$ series. An exception

Table 4.7. The 3d ions. m_{eff} is in units of μ_B

3d ⁿ		<i>S</i>	<i>L</i>	<i>J</i>	<i>g</i>	$m_{\text{eff}} = \frac{g\sqrt{J(J+1)}}{g\sqrt{J(J+1)}}$	$m_{\text{eff}} = \frac{g\sqrt{S(S+1)}}{g\sqrt{S(S+1)}}$	$m_{\text{eff}}^{\text{exp}}$
1	Ti ³⁺ , V ⁴⁺	$\frac{1}{2}$	2	$\frac{3}{2}$	$\frac{4}{5}$	1.55	1.73	1.7
2	Ti ²⁺ , V ³⁺	1	3	2	$\frac{2}{3}$	1.63	2.83	2.8
3	V ²⁺ , Cr ³⁺	$\frac{3}{2}$	3	$\frac{3}{2}$	$\frac{2}{5}$	0.78	3.87	3.8
4	Cr ²⁺ , Mn ³⁺	2	2	0	0		4.90	4.9
5	Mn ²⁺ , Fe ³⁺	$\frac{5}{2}$	0	$\frac{5}{2}$	2	5.92	5.92	5.9
6	Fe ²⁺ , Co ³⁺	2	2	4	$\frac{3}{2}$	6.71	4.90	5.4
7	Co ²⁺ , Ni ³⁺	$\frac{3}{2}$	3	$\frac{9}{2}$	$\frac{4}{3}$	6.63	3.87	4.8
8	Ni ³⁺	1	3	4	$\frac{5}{4}$	5.59	2.83	3.2
9	Cu ²⁺	$\frac{1}{2}$	2	$\frac{5}{2}$	$\frac{6}{5}$	3.55	1.73	1.9

seems to be Co²⁺, where an orbital contribution raises m_{eff} significantly above the spin only value.

Summarizing, for free ions:

- Filled electronic shells are not magnetic. A $\uparrow$ electron is paired with a $\downarrow$ electron in each orbital.
- Only partly filled shells may possess a magnetic moment.
- The magnetic moment is related to the angular momentum by $m = -g(\mu_B/\hbar)J$, where the quantum number J denotes the total angular momentum. For a given configuration, the values of J and g follow from Hund's rules and (4.10).

The third point in the summary has to be modified for ions in solids.

- Orbital angular momentum for 3d ions is quenched. The magnitude of the spin-only moment is $m \approx g\mu_B S$, with $g = 2$.
- Certain crystallographic directions become *easy axes* of magnetization.

These two effects result from electrostatic fields present in the crystal.

4.4.1 Crystal fields

When an ion or atom is embedded in a solid, the Coulomb interaction of its electronic charge distribution $\rho_0(\mathbf{r})$ with the surrounding charges in the crystal must be considered. This is the crystal-field interaction. Both the quenching of orbital angular momentum and the development of single-ion anisotropy are due to the crystal electric field.

The potential $\varphi_{cf}(\mathbf{r})$ produced by the distribution of charge $\rho(\mathbf{r}')$ outside the ion is

$$\varphi_{cf}(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}'|} d^3r'. \quad (4.27)$$

Table 4.8. Typical magnitudes of energy terms (in K) for 3d and 4f ions in solids

	$\mathcal{H}_0$	$\mathcal{H}_{so}$	$\mathcal{H}_{cf}$	$\mathcal{H}_Z$
3d	$1-5 \times 10^4$	10^2-10^3	10^4-10^5	1
4f	$1-6 \times 10^5$	$1-5 \times 10^3$	$\approx 3 \times 10^2$	1

$\mathcal{H}_Z$ is for 1 T

Here $1/|\mathbf{r} - \mathbf{r}'|$ can be expanded in spherical harmonics using spherical polar coordinates $\mathbf{r} = (r, \theta, \phi)$ and $\mathbf{r}' = (r', \theta', \phi')$:

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r'} \sum_{n=0}^{\infty} \frac{4\pi}{(2n+1)} \left(\frac{r}{r'}\right)^n \sum_{m=-n}^n (-1)^m Y_n^{-m}(\theta', \phi') Y_n^m(\theta, \phi).$$

Hence

$$\varphi_{cf}(\mathbf{r}, \theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n r^n \gamma_{nm} Y_n^m(\theta, \phi), \quad (4.28)$$

where

$$\gamma_{nm} = \frac{4\pi}{(2n+1)} \int \frac{\rho(\mathbf{r}')(-1)^m Y_n^{-m}(\theta', \phi')}{r'^{n+1}} d^3 r'.$$

The complete Hamiltonian of an ion in a solid has four terms:

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{so} + \mathcal{H}_{cf} + \mathcal{H}_Z.$$

$\mathcal{H}_0$ takes account of the Coulomb interactions among the electrons and between the electrons and the nucleus, giving rise to the total spin and orbital angular momenta $\mathbf{L}$ and $\mathbf{S}$. The terms $\mathcal{H}_{so}$, $\mathcal{H}_{cf}$ and $\mathcal{H}_Z$ are the spin-orbit, crystal-field and Zeeman terms, respectively. The crystal-field Hamiltonian is

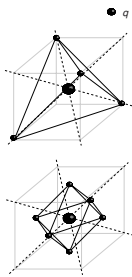
$$\mathcal{H}_{cf} = \int \rho_0(\mathbf{r}) \varphi_{cf}(\mathbf{r}) d^3 r. \quad (4.29)$$

Relative magnitudes of these interactions are given in Table 4.8.

$\mathcal{H}_{cf}$ is relatively weak in the rare-earths because the 4f shell is buried deep inside the atom (Fig. 3.12) and the crystal-field potential φ_{cf} is shielded by the outer electrons. $\mathcal{H}_{so}$ must be considered before $\mathcal{H}_{cf}$ in any perturbation scheme for evaluating the energies of 4f ions. J is a good quantum number, and the $|J, M_J\rangle$ -states form a basis. The crystal field is treated as a perturbation. The converse is true for 3d ions, where the 3d shell is outermost. The crystal field acts on the states $|L, M_L, S, M_S\rangle$ resulting from the Hamiltonian $\mathcal{H}_0$ for the 3d transition metal series. The orbital ground states for the 3d ions can only have $L = 0, 2$ or 3, that is an S, D or F state (Fig. 4.4(a)).

4.4.2 One-electron states

Generally, 4f electrons are localized in solids of any description, and 3d electrons are delocalized in metals, but usually localized in oxides and other ionic



Tetrahedral and octahedral coordination. The central on site has cubic symmetry in each case.

compounds, where their most common coordination is six-fold (octahedral) or four-fold (tetrahedral). If undistorted, both of these sites have cubic symmetry. The essential feature of cubic symmetry is four three-fold symmetry axes oriented along the cube's diagonals.

We begin by considering the effect of the crystal field on one-electron p - and d -states, and generalize to multielectron $3d$ ions in the next section. To demonstrate quenching of orbital angular momentum, we take the example of the p -states with $\ell = 1$ and $m_\ell = 0, \pm 1$. These orbital states, obtained from (4.1), and illustrated in Fig. 4.2, are

$$\begin{aligned}\psi_0 &= |0\rangle = R(r) \cos \theta, \\ \psi_{\pm 1} &= |\pm 1\rangle = (1/\sqrt{2})R(r) \sin \theta \exp(\pm i\phi).\end{aligned}$$

Next we suppose that a p^1 ion is surrounded by six other ions that form an octahedron. These could be oxygen O^{2-} anions. In the point-charge approximation the anions may be represented by point charges q at the corners of the octahedron. The crystal-field Hamiltonian $\mathcal{H}_{cf}$ depends on the Coulomb potential

$$V_{cf} = D_c(x^4 + y^4 + z^4 - 3y^2z^2 - 3z^2x^2 - 3x^2y^2), \quad (4.30)$$

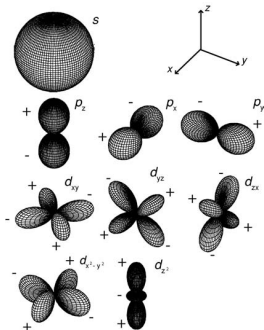
where $D_c = 7q/8\pi\epsilon_0 a^5$ and the six charges are placed at $\pm a$ along each of the x , y and z axes. However, the orbital states $|\pm 1\rangle$ are *not* eigenstates of $\mathcal{H}_{cf}$. In other words, the matrix elements $\langle i | \mathcal{H}_{cf} | j \rangle \approx eV_{cf}\delta_{ij}$ (δ_{ij} is the Kronecker delta-function; $\delta_{ij} = 0$ for $i \neq j$ and $\delta_{ii} = 1$); the 3×3 matrix which represents the crystal-field term in the Hamiltonian is not diagonal. This can be easily seen by inspecting the integrals over the wave functions, which are the matrix elements $\langle i | eV_{cf} | j \rangle = \int \psi_i^* eV_{cf} \psi_j d^3r$. The integral for $i = 1, j = -1$, for instance, involves $\sin^2\theta$ which is even in θ and will not integrate to zero. We need to find linear combinations of the hydrogenic orbital states that are eigenfunctions of $\mathcal{H}_{cf}$, namely

$$\begin{aligned}\psi_0 &= R(r) \cos \theta &= zR(r)/r &= p_z, \\ (1/\sqrt{2})(\psi_1 + \psi_{-1}) &= R(r) \sin \theta \cos \phi &= xR(r)/r &= p_x, \\ (i/\sqrt{2})(\psi_1 - \psi_{-1}) &= R(r) \sin \theta \sin \phi &= yR(r)/r &= p_y.\end{aligned}$$

These new wave functions are the p_x , p_y and p_z orbitals shown in Fig. 4.11. The expectation value of the z -component of angular momentum; $\ell_z = -i\hbar\partial/\partial\phi$ is zero for all three wave functions. The orbital angular momentum is said to be *quenched* by the crystal field. Unlike the rings of charge density which allow for clockwise or anticlockwise circulation of electrons in the xy -plane in ψ_1 and ψ_{-1} orbits of the free ion, no such circulation is possible in the p_x or p_y orbitals in the solid.

Figure 4.11

The s -, p - and d -orbitals, represented by boundary surfaces showing the angular distribution probability for electrons in each orbital. The sign of each wave function is indicated.



Similar arguments apply for the $3d$ -orbitals; the eigenfunctions there are three t_{2g} , and two e_g , orbitals⁴

t_{2g} orbitals:

$$d_{xy} = -(i/\sqrt{2})(\psi_2 - \psi_{-2}) = R'(r) \sin^2 \theta \sin 2\phi \approx xyR(r)/r^2$$

$$d_{yz} = (i/\sqrt{2})(\psi_1 + \psi_{-1}) = R'(r) \sin \theta \cos \theta \sin \phi \approx yzR(r)/r^2$$

$$d_{zx} = -(1/\sqrt{2})(\psi_1 - \psi_{-1}) = R'(r) \sin \theta \cos \theta \cos \phi \approx zxR(r)/r^2$$

e_g orbitals:

$$d_{x^2-y^2} = (1/\sqrt{2})(\psi_2 + \psi_{-2}) = R'(r) \sin^2 \theta \cos 2\phi \approx (x^2 - y^2)R(r)/r^2$$

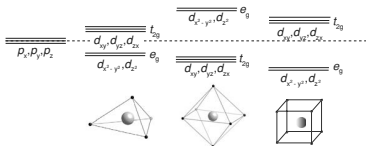
$$d_{3z^2-r^2} = \psi_0 = R(r)(3 \cos^2 \theta - 1) \approx (3z^2 - r^2)R(r)/r^2$$

Crystal fields and ligand fields The splitting of the $3d$ -orbitals in octahedral oxygen coordination, for example, is regarded in crystal-field theory as an electrostatic effect of the neighbouring oxygen anions, which are treated as point charges. This is an oversimplification. The $3d$ - and oxygen $2p$ -orbitals overlap, and they form a partially covalent bond. The oxygens bonding to the

⁴ *Notation:* Conventionally, a and b denote nondegenerate electron orbitals, e a two-fold degenerate orbital and t a threefold degenerate orbital. Lower-case letters refer to single-electron states, capital letters refer to multielectron states. a , A are nondegenerate and symmetric with respect to the principal axis of symmetry (the sign of the wave function is unchanged); b , B are antisymmetric with respect to the principal axis (the sign of the wave function changes). Subscripts g and u indicate whether the wave function is symmetric or antisymmetric under inversion. Subscript 1 refers to mirror planes parallel to a symmetry axis, 2 refers to diagonal mirror planes.

Figure 4.12

Splitting of one-electron $3d$ energy levels in different cubic crystal fields; $2p$ levels are unsplit.

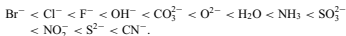


$3d$ metal are the ligands. On account of their orientation, the overlap is greater for e_g - than t_{2g} -orbitals. The overlap leads to mixed wave functions, producing bonding and antibonding orbitals, whose splitting increases with overlap. The hybridized orbitals are

$$\phi = \alpha \psi_{2p} + \beta \psi_{3d},$$

where $\alpha^2 + \beta^2 = 1$, for normalization. The splitting between the mainly $t_{2g} \pi^*$ and the mainly $e_g \sigma^*$ -orbitals for $3d$ ions in octahedral coordination⁵ is usually 1–2 eV. The ionic and covalent contributions to the splitting Δ are actually of comparable magnitude.

The sequence of ligands in order of increasing effectiveness at producing crystal/ligand field splitting Δ is known as the spectrochemical series. The ligands that concern us here, in a sequence which reflects both ionic charge and the tendency to bond covalently are



The bond is mostly ionic at the beginning of the series, and mostly covalent at the end.

Covalency is stronger in tetrahedral coordination, on account of the shorter metal-oxygen distances, but the crystal-field splitting Δ_{tet} is $\frac{4}{9} \Delta_{oct}$. The relative splittings of the $3d$ crystal-field levels in different sites with cubic symmetry are illustrated in Fig. 4.12 for the one-electron states. The symmetry of the one-electron states is preserved even in metals, where the coordination number is 8 or 12, and the neighbours appear negatively charged.

As the site symmetry is reduced, the degeneracy of the one-electron energy levels is raised. For example, a tetragonal extension of the octahedron along the

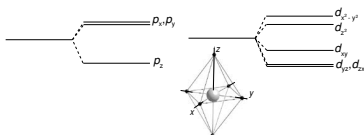
⁵ σ -bonds have charge density mainly around the line joining the atoms, whereas π -bonds have charge density above or below this line. The * denotes an antibonding orbital. The $3d$ -states usually lie higher in energy than the $2p$ -states in oxides, so the hybridized orbitals with mainly $2p$ character are bonding and those with mainly $3d$ character are antibonding.

Table 4.9. The splitting of the one-electron levels in different symmetry

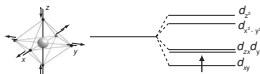
	ℓ	Cubic	Tetragonal	Trigonal	Rhombohedral
s	0	1			
p	1	3	1, 2	1, 2	1, 1, 1
d	2	2, 3	1, 1, 1, 2	1, 2, 2	1, 1, 1, 1, 1
f	3	1, 3, 3	1, 1, 1, 2, 2	1, 1, 1, 2, 2	1, 1, 1, 1, 1, 1, 1

Figure 4.13

The effect of a tetragonal distortion of octahedral symmetry on the one-electron energy levels.

**Figure 4.14**

Jahn-Teller distortion of an octahedral site containing a d^1 ion.

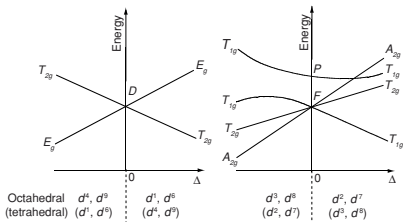


z axis will lower p_z and raise p_x and p_y . The effect on the d -states is shown in Fig. 4.13. The degeneracy of the d -orbitals in different symmetries is shown in Table 4.9.

A system with a single electron (or hole) in a degenerate level will tend to distort spontaneously as shown in Fig. 4.14. This is the Jahn-Teller effect. The effect is particularly strong for d^4 and d^9 ions in octahedral symmetry (Mn^{3+} , Cu^{2+}), which can lower their energy by distorting the crystal environment. If the local strain is ϵ , the energy change $\delta\epsilon = -A\epsilon + B\epsilon^2$, where the first term is the crystal-field stabilization energy and the second term is the increased elastic energy. Here A and B are constants. The minimum is at $\epsilon = A/2B$. The crystal field plays an important part in stabilizing ionic structures. The stabilization energy for a d^1 ion is $0.4\Delta_{\text{oct}}$ in an octahedral site, and it can be further increased by a uniaxial distortion that splits the t_{2g} levels. The splittings conserve the centre of gravity of the levels, so the e_g levels are raised by $0.6\Delta_{\text{oct}}$ while the t_{2g} levels are lowered by $-0.4\Delta_{\text{oct}}$ in an octahedral crystal field. The d^5 ions Fe^{2+} or Mn^{2+} , which have all orbits singly occupied, exhibit no crystal-field stabilization energy.

Figure 4.15

Orgel diagrams for D and F terms. The corresponding d^n ion configuration is shown.



4.4.3 Many-electron states

The one-electron picture is modified by the strong Coulomb interactions among the electrons. There are only three distinct types of energy-level diagrams for $3d$ ions, corresponding to the D , F or S ground-state terms. The D -states map directly onto the one-electron levels because there is just one electron or hole in an otherwise empty, half-filled or filled d -shell. These are the d^1 , d^4 , d^6 and d^9 configurations, with $L = 2$ according to Hund's second rule. The F term is found for the d^2 , d^3 , d^7 and d^8 configurations, with $L = 3$. The S term is for d^5 , where $L = 0$. Orgel diagrams (Fig. 4.15) show the splitting of the ground-state D or F level in a cubic crystal field. The S term does not split, on account of the spherical symmetry of the half-filled shell.

The hierarchy of interactions $\mathcal{H}_0 > \mathcal{H}_{cf} > \mathcal{H}_{so}$ is not always maintained for $3d$ ions. Some ligands create such intense electrostatic fields that they manage to overturn not only Hund's second rule, but also his first rule as well. The crystal field can drive the ion into a low-spin state. The on-site Coulomb interaction has the effect of raising the energy of doubly occupied orbitals by U , compared with singly occupied orbitals. This is the electrostatic penalty for double occupancy. When U exceeds the crystal-field splitting of the one-electron levels, Hund's first rule applies as advertised, but when $U < \Delta$ the t_{2g} -orbitals for an octahedral site, for example, will tend to be doubly occupied. Figure 4.16 illustrates the high-spin and low-spin states for Fe^{2+} .

In some materials, where the low-spin state lies only slightly lower in energy than the high-spin state, a spin crossover may occur as a function of temperature at a phase transition, driven by magnetic entropy $R \ln(2S + 1)$.

Tanabe-Sugano diagrams show the splitting of both ground-state and higher terms in the crystal field. They are drawn with the ground-state energy set to zero. Some diagrams for $3d$ ions in octahedral coordination are shown in Fig. 4.17. The stabilization of low-spin states in strong crystal fields is evident

Figure 4.16

Comparison of the one-electron energy levels for Fe^{2+} ($3d^6$) in FeCl_2 , where there is a high-spin state with $S = 2$, and FeS_2 where the crystal field stabilizes a low-spin state with $S = 0$.

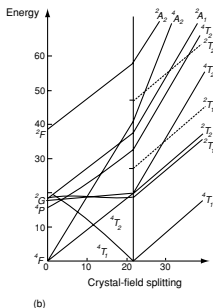
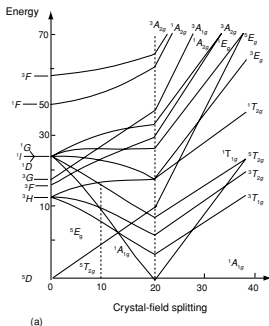
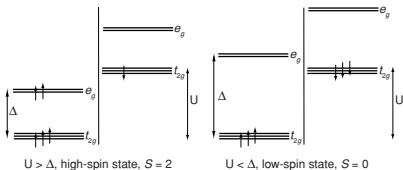


Figure 4.17

Tanabe-Sugano diagrams for: (a) a $3d^5$ ion and (b) a $3d^7$ ion. The high-spin-low-spin crossover is indicated by the vertical line.

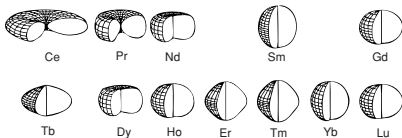
from these diagrams. The crystal-field parameters, and especially the cubic crystal-field parameter Δ (also known as $10Dq$) can be deduced from the wavelengths of optical transitions from the ground state to different excited states.

4.4.4 Single-ion anisotropy

Single-ion anisotropy is due to the electrostatic interaction (4.31) of charge in orbitals containing the magnetic electron distribution $\rho_0(\mathbf{r})$ with the potential $\varphi_{cf}(\mathbf{r})$ created at the atomic site by the rest of the crystal. As we have seen in Figs. 4.14–4.17, the crystal-field interaction tends to stabilize particular orbitals.

Figure 4.18

Charge density distributions for the trivalent rare-earth ions at zero temperature. The deviations from spherical symmetry are exaggerated.



Spin-orbit interaction $\Lambda \mathbf{L} \cdot \mathbf{S}$, then leads to magnetic moment alignment in specific directions in the crystal.

Crystal-field anisotropy is easiest to treat in the case of the rare-earths. With these ions it is possible to cleanly divide the charge density $e\rho(\mathbf{r}')$ producing the crystal field, from the charge $e\rho_{4f}(\mathbf{r})$ in the $4f$ shell on which it acts. If the electrostatic potential due to the surroundings is $\varphi_{cf}(\mathbf{r})$ and the $4f$ electron density is ρ_{4f} , the interaction energy is

$$\varepsilon_a = \int e\rho_{4f}(\mathbf{r})\varphi_{cf}(\mathbf{r})d^3r. \quad (4.31)$$

It is appropriate to expand the $4f$ charge density in spherical harmonics and express the electrostatic interaction in terms of the 2^n -pole moments of the charge distribution, where n is even. The $n=2$ term is the quadrupole moment:

$$Q_2 = \int \rho_{4f}(r)(3\cos^2\theta - 1)r^2d^3r.$$

The sign of Q_2 reflects the shape of the $4f$ electron cloud whether prolate (elongated, $Q_2 > 0$) or oblate (flattened, $Q_2 < 0$), as shown in Fig. 4.18. Units are m^2 .

The $n=4$ and $n=6$, hexadecapole and 64-pole moments are defined as

$$Q_4 = \int \rho_{4f}(r)(35\cos^4\theta - 30\cos^2\theta + 3)r^4d^3r,$$

$$Q_6 = \int \rho_{4f}(r)(231\cos^6\theta - 315\cos^4\theta + 105\cos^2\theta - 5)r^6d^3r.$$

These higher-order moments decide, for instance, whether a prolate charge distribution resembles a bone ($Q_4 < 0$) or a snake which has just eaten a rabbit ($Q_4 > 0$). Units are m^4 and m^6 , respectively.

In a few cases it is possible to represent the multipole moments Q_m in terms of closed expressions. For example, in the $J = J_z$ ground state of the heavy rare-earths where $\hat{\mathbf{O}}_2^0 = 2J^2 - J$, quadrupole moments are given by

$$Q_2 = -(1/45)(14 - n)(7 - n)(21 - 2n)\langle r_{4f}^2 \rangle,$$

where n is the number of $4f$ electrons. The leading term in (4.31) is

$$\varepsilon_a = (1/2)Q_2A_2^0(3\cos^2\theta - 1), \quad (4.32)$$

where A_2^0 is the second-order uniaxial crystal-field parameter that describes the electric field gradient created by the charge distribution of the crystal at the rare-earth atomic site, which interacts with the electric quadrupole moment Q_2 . Hence $A_2^0 = -(e/16\pi\epsilon_0) \int [(3r_z'^2 - r'^2)/r'^5] \rho(r') d^3r'$. Stevens showed that the crystal-field interaction can be expressed in terms of the angular momentum operators (an example of the Wigner-Eckart theorem). He wrote the crystal-field Hamiltonian as

$$\mathcal{H}_{cf} = \sum_{n=0,2,4,6} \sum_{m=-n,\dots,n} B_n^m \hat{\mathbf{O}}_n^m, \quad (4.33)$$

where $B_n^m = \theta_n \langle r_{4f}^n \rangle A_n^m$, and θ_n is a constant, different for each rare-earth, which is proportional to the 2^n -pole moment: $Q_2 = 2\theta_2 \langle r_{4f}^2 \rangle$, and $Q_4 = 8\theta_4 \langle r_{4f}^4 \rangle$, $Q_6 = 16\theta_6 \langle r_{4f}^6 \rangle$, and $\hat{\mathbf{O}}_n^m$ are the Stevens operators, tabulated in Appendix H and by Hutchings (1965). For example,

$$\hat{\mathbf{O}}_2^0 = [3\hat{J}_z^2 - J(J+1)].$$

Here we have dropped the $\hbar$ in the angular momentum operators. The temperature dependence of the multipole moments Q_n can be evaluated from the thermodynamic averages $\langle \hat{\mathbf{O}}_n^0 \rangle$.

In rare-earths, a limited number of terms in the expansion (4.33) are needed, depending on the symmetry of the site. The crystal-field coefficients B_n^m or A_n^m are then parameters to be determined experimentally. At low temperatures, second-, fourth- and sixth-order terms may all be important, but at room temperature it is often enough to consider only the leading, second-order terms

$$\mathcal{H}_{cf} = \theta_2 \langle r_{4f}^2 \rangle [A_2^0 \hat{\mathbf{O}}_2^0 + A_2^2 \hat{\mathbf{O}}_2^2], \quad (4.34)$$

which, in the case of axial symmetry when there is no off-diagonal term, is further simplified to

$$\mathcal{H}_{cf} = D \hat{J}_z^2, \quad (4.35)$$

where D is the uniaxial crystal field parameter.

Another commonly encountered expression, for cubic anisotropy, is

$$\mathcal{H}_{cf} = \theta_4 \langle r_{4f}^4 \rangle [A_4^0 \hat{\mathbf{O}}_4^0 + 5A_4^4 \hat{\mathbf{O}}_4^4] + \theta_6 \langle r_{4f}^6 \rangle [A_6^0 \hat{\mathbf{O}}_6^0 - 21A_6^4 \hat{\mathbf{O}}_6^4]. \quad (4.36)$$

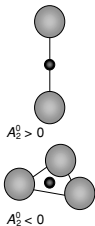
Only the fourth-order term exists for $3d$ ions, with r_{3d} taking the place of r_{4f} .

A general expression for ε_a is given in Appendix H.

Kramer's theorem, which is a consequence of time reversal symmetry, states that energy levels for systems with an odd number of electrons, and therefore half-integral quantum numbers, are always at least two-fold degenerate in the absence of a magnetic field. When J is half-integral, the second-order crystal field produces a series of *Kramer's doublets* $|\pm M_J\rangle$; but when J is integral there is a singlet $|0\rangle$ and a series of doublets. If D is negative, the $|\pm J\rangle$ state is

Table 4.10. Data on the rare-earth ions. The operators $\hat{O}_n^0$ are evaluated at $T = 0\text{ K}$, where $J_z = J$

	J	θ_2 (10^{-2})	$\theta_2 \langle r^2 \rangle O_2^0$ (a_0^2)	θ_4 (10^{-4})	$\theta_4 \langle r^4 \rangle O_4^0$ (a_0^4)	θ_6 (10^{-6})	$\theta_6 \langle r^6 \rangle O_6^0$ (a_0^6)	G	γ_s	C_{mol}
Ce ³⁺	5/2	-5.714	-0.748	63.5	1.51			0.18	-1/3	8.0
Pr ³⁺	4	-2.101	-0.713	-7.346	-2.12	60.99	5.89	0.80	-1/2	16.0
Nd ³⁺	9/2	-0.643	-0.258	-2.911	-1.28	-37.99	-8.63	1.84	-1/4	16.4
Sm ³⁺	5/2	4.127	0.398	25.012	0.34			4.46	-5	0.9
Gd ³⁺	7/2				-			15.75	1	78.8
Tb ³⁺	6	-1.010	-0.548	1.224	1.20	-1.12	-1.28	10.50	2/3	118.2
Dy ³⁺	15/2	-0.635	-0.521	-0.592	-1.46	1.04	5.64	7.08	1/2	141.7
Ho ³⁺	8	-0.222	-0.199	-0.333	-1.00	-1.29	-10.03	4.50	2/5	140.7
Er ³⁺	15/2	0.254	0.190	0.444	0.92	2.07	8.98	2.55	1/3	114.8
Tm ³⁺	6	1.010	0.454	1.633	1.14	-5.61	-4.05	1.17	2/7	71.5
Yb ³⁺	7/2	3.175	0.435	-17.316	-0.79	1.48	0.73	0.29	1/4	25.7



Examples of atomic configurations that produce positive and negative electric field gradients at the central site.

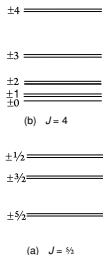
lowest, and the susceptibility in a magnetic field applied along the crystal-field axis is given by (4.14). If D is positive, the $M_J = 0$ singlet is the ground state. The crystal field appears to have destroyed the moment. The initial susceptibility is zero, although the induced moment increases quadratically with field. However, when the magnetic field is applied perpendicular to the crystal-field axis, there is a large susceptibility. What this means is that there is an easy axis of magnetization in the first case, but an easy plane in the second.

Single-ion anisotropy is the major source of anisotropy in hard ferromagnetic materials. The tendency for the magnetic moment to lie along specific crystal axes generally makes the susceptibility a tensor rather than a scalar quantity. There are three principal axes, for which the induced moment lies parallel to the applied magnetic field.

In general, the details of the response of the system to an applied field are obtained by diagonalizing the Hamiltonian $\mathcal{H}_{cf} + \mathcal{H}_z$, and evaluating $\langle m \rangle$ from (4.11). In a series of isostructural rare-earth metals or compounds, A_n^m is roughly constant. The leading term in the anisotropy just depends on the product of A_2^0 and $\theta_2 \langle r^2 \rangle J_z^2$ for the rare-earth in question. Both θ_2 , and the quadrupole moment change sign at each quarter-shell filling (Fig. 4.18, Table 4.10). When A_2^0 is positive, ions such as Nd³⁺, which have a negative quadrupole moment and an oblate charge distribution, will exhibit easy axis anisotropy, whereas ions such as Sm³⁺, which have a positive quadrupole moment and a prolate charge distribution, will exhibit hard axis anisotropy.

Consider a rare-earth, for which the spin-orbit interaction stabilizes a $(2J + 1)$ -fold degenerate multiplet $|J, M_J\rangle$. The crystal field defines the z axis and splits out the M_J states, subject to Kramers theorem.

In the figure, Sm³⁺ has a $|\pm 5/2\rangle$ ground state, so the moment has its maximum positive or negative projection along Oz , which precesses as indicated by the vector model. The average moment lies along the z -direction, and



Crystal field splitting for ions in an axial field with $A_2^0 < 0$: (a) Sm^{3+} ; (b) Pr^{3+} .

application of a field in this direction splits out the $\pm \frac{5}{2}$ states, leading to a susceptibility $\chi = C/T$ given by (4.14) with $J = \frac{5}{2}$. When the magnetic field is applied in the x -direction, the susceptibility is much smaller.

The situation is different for Pr^{3+} . The crystal field in this case makes $|0\rangle$ the ground state, with no projection of the moment along Oz . The ground state is nonmagnetic in the vector model, the moment in the $M_J = 0$ state lies in an indeterminate direction in the xy -plane. The susceptibility is zero at low temperature in a small magnetic field applied along Oz . However, in large fields, the Zeeman splitting of the excited states $|\pm 1\rangle$, $|\pm 2\rangle$, $|\pm 3\rangle$ and $|\pm 4\rangle$ produces a series of level crossings and the magnetization in high fields is ultimately that of the $|\pm 4\rangle$ state. However, if the field is applied along Ox , the situation is quite different. There is immediately a high susceptibility associated with the $|\pm 4\rangle$ state. The Pr^{3+} in this example has hard axis-easy plane anisotropy.

Data on the rare-earth ions are collected in Table 4.10. The fourth, sixth and eighth columns respectively indicate the relative strengths of the second-, fourth- and sixth-order crystal-field interactions; G is the de Gennes factor, γ_s is the ratio of spin to total moment and C_{mol} is the molar Curie constant. The crystal-field parameters A_n^m or B_n^m are determined by experiment.

4.4.5 The spin Hamiltonian

The orbital states of a multielectron $3d$ ion are represented by the states $|L, M_L\rangle$ and the spin states by $|S, M_S\rangle$. Since the orbital moments are quenched by the crystal field, the effects of the crystal field and spin-orbit coupling on the magnetic energy levels of an ion are conveniently represented by an effective spin Hamiltonian based on the $2S + 1$ $|M_S\rangle$ sublevels of the ground state. The magnetic properties of the ion depend on the splitting of these levels in a magnetic field.

Following (4.35), the simplest spin Hamiltonian, used for sites with uniaxial symmetry, is

$$\mathcal{H}_{spin} = D S_z^2. \quad (4.37)$$

An orthorhombic distortion adds a term $E(S_x^2 - S_y^2)$. The next term in tetragonal symmetry is $F S_z^4$. Cubic symmetry gives a term $D_c(S_x^4 + S_y^4 + S_z^4)$. The splitting of the levels usually depends on the orientation of the applied field, which is reproduced by an anisotropic $\hat{g}$ -tensor, so the Zeeman term is written as $\mu_B/\hbar (\mathbf{B} \cdot \hat{g} \cdot \mathbf{S}) = (\mu_B/\hbar) \sum_{i,j} g_{ij} B_i S_j$. The spin Hamiltonian approach is widely used in electron paramagnetic resonance, discussed in Chapter 9, where the spectra are parameterized in terms of D , E , $\dots$, $g_{\parallel}$, $g_{\perp}$. The ability to diagonalize large Hamiltonian matrices which include crystal-field, spin-orbit and Zeeman terms by computer has reduced the need for these approximation methods.

FURTHER READING

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EXERCISES

- 4.1 Calculate the multiplet splitting in terms of the spin-orbit coupling constant Λ for an ion with $L = 3$, $S = \frac{1}{2}$.
- 4.2 Deduce the expression for the Langevin function from (4.20).
- 4.3 Calculate the Van Vleck susceptibility for a mole of Eu^{3+} ions.
- 4.4 Plot the crystal field stabilization energy for high-spin $3d^n$ ions for $1 \leq n \leq 9$ in (a) octahedral and (b) tetrahedral sites. Use the one-electron levels shown in Fig. 4.12 and take $\Delta_{tet} = -(4/9)\Delta_{oct}$.
- 4.5 Make a table of the spin moments of $3d$ ions in six-fold, octahedral coordination in high- and low-spin states. Repeat for eight-fold cubic coordination and four-fold tetrahedral coordination.
- 4.6 At zero temperature in a compound containing octahedrally coordinated Fe^{2+} , the low-spin energy level lies just 50 meV below the high-spin energy level. At what temperature would you expect spin crossover to occur? What is the order of the transition?
- 4.7 Show that the low-temperature specific heat is linear in temperature for a sample containing Kramers ions which are subject to a uniform magnetic field when the local crystal field axes are oriented at random with respect to the field direction.
- 4.8 An ion is in a site where the spin Hamiltonian is DS_z^2 . If $D = 6$ K, $S = 1$ and $g = 2$. Sketch the magnetization as a function of field when the field is applied (a) along Oz and (b) along Ox at 0 K and 4 K. Make use of the angular momentum operators for $l = 1$, given in §3.1.4.
- 4.9 Show that for an ion in an $S = 1$ state, described by the spin Hamiltonian $\mathcal{H} = g_{\parallel}\mu_B B_z S_z + g_{\perp}\mu_B (B_x S_x + B_y S_y) + DS_z^2$, the energy eigenstate ε can be deduced from $\varepsilon = (\varepsilon - D)^2 - \varepsilon(g_{\parallel}\mu_B B \cos \theta)^2 - (\varepsilon - D)(g_{\perp}\mu_B \sin \theta)^2$. Find the solutions for $\theta = 0$ and $\theta = \pi/2$. Make use of the angular momentum operators for $L = 1$.